

Envoy — Submission Package

Copy-paste text for the form. Use this at ~5 PM to submit fast.

Voice Agents Hackathon (Cekura × Daily × NVIDIA × AWS), May 30 · Doc 4 of 4

Required fields

Team / project name

Envoy

Team members

- Rayan Arya — [email]
- Rushil [last name] — [email]

Git URL

[https://github.com/\[org\]/envoy](https://github.com/[org]/envoy) — README links to the live demo + the video.

Submission video URL

Upload to YouTube (unlisted) or similar durable host. Test in incognito before pasting. [URL]

Live “try it” link (if shipped)

[URL]

Project description

Short (1–2 sentences — one-line fields)

Envoy is a voice-first autonomous agent that gets things done for you over the phone — it calls real businesses, negotiates, books, and writes the result into your real tools — and can't be manipulated into betraying you. When a new attack slips through, it writes its own guardrail, verifies the fix, and never falls for it again.

Medium (paragraph — description fields)

Envoy is a voice-first autonomous agent built on the full hackathon stack. You speak a goal; it plans the task, places a real phone call (to a human business or an AI agent), negotiates within your constraints, books, and writes a real Google Calendar event + confirmation email. Because it acts on your behalf with your data and money, it treats every counterpart as untrusted: it blocks manipulation in real time, and when a novel attack slips through, it diagnoses it, writes a new guardrail, verifies the fix against the attack and its variants, and stores the hardened version — a self-improving agent that measurably gets stronger over time. The reasoning brain is an open-weights model served on NVIDIA NIM + AWS GPU with NVIDIA Riva speech; the pipeline runs on Pipecat with Twilio telephony; Cekura powers the adversarial evaluation and the auto-improve loop.

Long (multi-paragraph — “tell us more” fields)

Today's assistants answer; they don't act. The blocker isn't intelligence — it's trust: nobody hands an agent their calendar, inbox, and budget if it can be talked into anything by whatever it bumps into. Envoy is the autonomous agent you can finally trust to act alone.

What it does: understands a spoken goal, places a real call, negotiates within constraints, books, and writes real side effects (Calendar event + email). **What makes it safe:** it holds your constraints and refuses to leak credentials, exceed budget, or obey injected instructions — from a human or an AI counterpart. **What makes it production-grade:** a continuous self-heal loop — Cekura-scored evaluation flags a failure, a Patcher writes a new guardrail, the identical attack is replayed and verified blocked, and a never-seen variant is blocked too, proving it learned the principle rather than memorizing the attack.

How it maps to the brief: Build & Customize — open-weights LLM (Llama/Nemotron) on NVIDIA NIM + AWS, policy-tuned, with Riva speech. Deploy at Scale — Pipecat pipeline over Twilio PSTN, streaming with sub-second turn latency. Simulate & Evaluate — Cekura adversarial + functional scenarios via MCP, LLM-judged per turn. Auto-Improve — the self-heal loop is the centerpiece: eval data flows back into the agent as new guardrails, versioned over time.

The wedge: high-stakes outbound tasks where a mistake is expensive and a manipulated agent is a liability. We think the trust layer is what every autonomous-action product will need within 18 months.

Sponsor / side-quest entries

NVIDIA — best use of NVIDIA

Envoy's reasoning brain is an open-weights model (Llama/Nemotron) served via NVIDIA NIM on an AWS GPU, policy-tuned for constraint-holding negotiation, and its ears + voice are NVIDIA Riva STT/TTS. NVIDIA accelerates both the high-reasoning brain and the real-time speech that makes sub-second phone conversations possible.

AWS — best use of AWS

AWS provides the GPU compute serving the open-weights brain, hosts the full pipeline + live link, and stores the guardrail-version history and evaluation records that form Envoy's "flywheel memory" — the record of every attack it has hardened against.

Cekura — best use of Cekura

Cekura is the scoring backbone of Envoy's auto-improve loop. We run adversarial + functional scenarios against Envoy via Cekura's MCP; its LLM-judge scores each turn into structured pass/fail + severity, which the Patcher turns into new guardrails. We're a power user: the eval engine is what makes "self-healing" measurable rather than vibes.

Daily / Pipecat — best build on the stack

Envoy's entire real-time voice pipeline is Pipecat — transport, streaming STT/TTS, barge-in, and the agent-to-agent call between Envoy and a counterpart agent all run on it.

Twilio — best use of Twilio

Envoy dials real phone numbers over Twilio PSTN — a human restaurant or an AI booking line — and runs the same pipeline either way, with the call recorded for the demo and a confirmation surfaced after booking.

README hero (top of envoy/README.md)

```
# Envoy

**Envoy is an autonomous voice agent that calls real businesses to get things done for you -
and hardens
itself against manipulation in real time.**

We built: (1) an outbound voice agent that places a real phone call, negotiates, and books;
(2) real side
effects (Google Calendar event + confirmation email); (3) a self-heal loop - when a
counterpart manipulates
the agent, it writes a new guardrail, replays the attack, and verifies it's blocked (incl.
reworded variants).

Voice-first autonomous agent. Calls real businesses (human or AI), negotiates, books, and
writes real
side effects (Calendar + email) - while treating every counterpart as untrusted. When a new
attack
slips through, it writes its own guardrail, verifies the fix, and never falls for it again.

Built for the Voice Agents Hackathon (Cekura x Daily x NVIDIA x AWS), May 30 2026.

## How it hits the brief's four stages
- Build & Customize - open-weights brain (Llama/Nemotron) on NVIDIA NIM + AWS GPU; Riva speech
- Deploy at Scale - Pipecat pipeline over Twilio PSTN; sub-second turn latency
- Simulate & Evaluate - Cekura adversarial + functional scenarios via MCP, LLM-judged
- Auto-Improve - the self-heal loop: flagged failure -> new guardrail -> replay -> verified
blocked

## Built by
- Rayan Arya
- Rushil [last name]

## Demo
[video link] ## Try it: [live link]

## Run
git clone https://github.com/[org]/envoy && cd envoy && (see SETUP.md for keys) && make dev
```

Pre-submit final checks (5 min, ~5:55 PM)

- Team/project name + both emails filled.
- Git URL points to envoy with a working README + architecture diagram.
- Video link accessible in incognito.
- Project description pasted; sponsor entries pasted in their fields.
- Repo public, not private; commit history clean; no secrets committed.

- Live link works (if shipped).
- Brief's four stages each named in the write-up.
- Demo video is 60 seconds or less (hard cap); the real call + the self-heal flash are both in it.
- Submission form filled with buffer before 6 PM.

The lines to leave them with — one-liner: “An AI that gets things done for you over the phone, and can’t be manipulated into betraying you.” Beat-2 caption: “Real restaurant. Real human. Real calendar. Not a mockup.” Beat-4 caption: “It didn’t memorize the attack — it learned the rule.” Close: “We built the one you can trust to act — alone, in a world full of agents that can’t be trusted.”